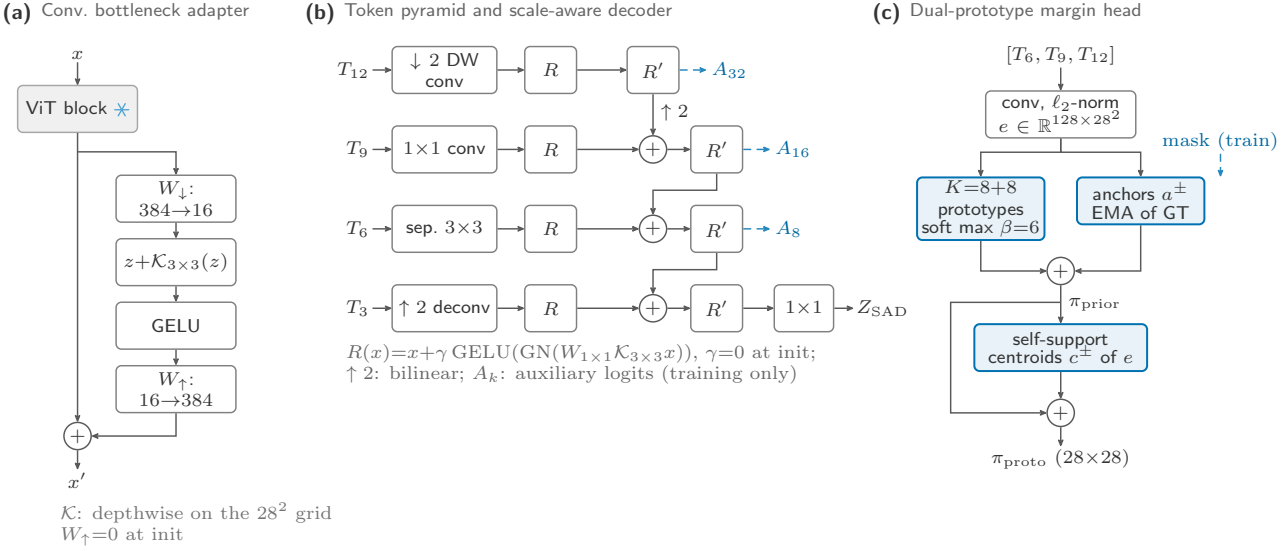

(a) Conv. bottleneck adapter
x
ViT block
$W_{\downarrow}$: 384→16
$z+\mathcal{K}_{3\times3}(z)$
GELU
$W_{\uparrow}$: 16→384
x'
$\mathcal{K}$: depthwise on the 28^2 grid
$W_{\uparrow}$=0 at init
(b) Token pyramid and scale-aware decoder
T_{12}
↓ 2 DW conv
R
R'
A_{32}
↑ 2
T_9
1×1 conv
A_{16}
T_6
sep. 3×3
A_8
T_3
↑ 2 deconv
1×1
Z_{SAD}
$R(x)=x+\gamma\,\mathrm{GELU}(\mathrm{GN}(W_{1\times1}\mathcal{K}_{3\times3}x))$, γ=0 at init;
↑ 2: bilinear; A_k: auxiliary logits (training only)
(c) Dual-prototype margin head
$[T_6, T_9, T_{12}]$
conv, ℓ_2-norm
$e \in \mathbb{R}^{128\times28^2}$
mask (train)
K=8+8 prototypes soft max β=6
anchors $a^{\pm}$ EMA of GT
π_{prior}
self-support centroids $c^{\pm}$ of e
π_{proto} (28×28)